TECHNICAL & OPERATIONAL REFERENCE

Simulation of Clinical Skills · Digital Handbook

Kabinet Vještina · University of Rijeka, Faculty of Medicine

STACK	DEPLOY	CONTENT	STATUS
Docusaurus v3 + React	**Netlify / GitHub Pages**	**MDX (Markdown + JSX)**	**v0.1 — Scaffold phase**

1. Project Overview

PURPOSE

The CSS Digital Handbook is a publicly hosted web application that reproduces the full content of "Simulation of Clinical Skills: A Practical Guide" (108 pages, 7 clinical parts) as a navigable, searchable, mobile-friendly reference site for sixth-year medical students at the University of Rijeka Faculty of Medicine.

The site serves as the primary digital companion to the Kabinet Vještina (Skills Cabinet) course, accessible year-round alongside the Merlin e-learning platform. It is not a replacement for Merlin — it is the permanent, open-access reference layer.

DESIGN REFERENCE

The visual identity is established by the landing page (landing.html). All component and theme decisions derive from it. The core aesthetic is: deep navy base, warm gold accent, editorial serif typography (Cormorant Garamond / Garamond for display, DM Mono for labels and code, DM Sans for body).

NON-GOALS

This system does NOT replace Merlin or the LMS.

It does NOT include student authentication, grade tracking, or submission workflows.

Interactive quizzes and algorithm simulations are a future phase — not in scope for v1.

It is NOT a visual replica of the printed book layout.

2. Technology Stack

2.1 Core Framework — Docusaurus v3

Docusaurus is a Meta-maintained static site generator purpose-built for documentation and reference sites. It outputs a React-based SPA with static pre-rendering, meaning fast load times, excellent SEO, and zero server infrastructure required.

Key reasons for choosing Docusaurus over alternatives:

- **MDX content:** Every page is a .mdx file (Markdown + React). Content is plain text, editable by non-developers, and git-versionable. Custom React components (clinical callout boxes, ABCDE templates, figure placeholders) can be dropped inline.

- **Built-in full-text search:** Algolia DocSearch (free for open-source/educational projects) or the local @docusaurus/plugin-search-local plugin provides instant search across all 108 pages of content.
- **Auto-generated sidebar:** The sidebar is automatically built from the file/folder structure. Adding a new chapter means creating a new .mdx file — no manual navigation config needed.
- **Zero-config deployment:** One command deploys to GitHub Pages (docusaurus deploy) or Netlify via CI/CD. No server, no database, no maintenance burden.
- **Extensible:** The swizzle system allows overriding any built-in component. The custom landing page (landing.html) becomes the Docusaurus homepage via a custom React component, preserving the full design.

Alternative	Why Rejected
MkDocs	Python-based, limited React extensibility, weaker MDX support
Plain HTML/CSS site	No search, no auto-nav, no versioning, high maintenance
Next.js	Overkill for static docs; requires more manual wiring
Notion / GitBook	No full custom design control; vendor lock-in

2.2 Content Format — MDX

Each handbook chapter is a .mdx file. MDX is a superset of Markdown that supports JSX/React component imports inline. This means:

- Standard Markdown for paragraphs, headings, tables, and lists
- Custom React components for clinical-specific UI (ABCDE boxes, key-point callouts, figure placeholders, algorithm cards)
- No HTML spaghetti — content stays readable and editable as plain text

2.3 Hosting & Deployment

Primary target: Netlify (recommended) or GitHub Pages. Both are free for public repositories.

Platform	Notes
Netlify	Recommended. Auto-deploys on every git

Platform	Notes
	push. Free SSL, custom domain, CDN. Deploy previews on PRs.
GitHub Pages	Alternative. Free. Slightly more config required for Docusaurus base URL.
Custom Faculty Server	Possible — output is a static build folder. Copy /build to any web server.

2.4 Search

Phase 1 uses @docusaurus/plugin-search-local — a fully offline, no-API-key search that indexes all content at build time. For Phase 2+, Algolia DocSearch can be applied for (free for educational projects) to get a faster, cloud-powered search experience.

2.5 Fonts

Fonts are loaded via Google Fonts CDN (as in the landing page) and referenced in the Docusaurus custom CSS:

```
/* src/css/custom.css */
@import url('https://fonts.googleapis.com/css2?
  family=Cormorant+Garamond:ital,wght@0,300;0,400;0,600;1,300;1,400
  &family=DM+Mono:wght@300;400;500
  &family=DM+Sans:wght@300;400;500&display=swap');
```

3. Design System

All design decisions are derived from the established landing page. This section is the canonical reference for any new component or page built in the project.

3.1 Colour Tokens

CSS Variable	Hex	Usage
--navy	#0B1120	Page background, nav, footer
--navy2	#111C30	Section alternate background
--navy3	#162038	Cards, elevated surfaces
--gold	#C8A96E	Primary accent, headings, borders
--gold2	#E2C98A	Hover states, lighter gold
--cream	#F2EDE4	Primary text on dark backgrounds
--cream2	#D8D0C4	Secondary / muted text
--muted	#7A8499	Labels, metadata, placeholder text

3.2 Typography Scale

Role	Font Family	Size / Weight	Usage
Display H1	Cormorant Garamond	clamp(3.5rem–7rem) / 300	Hero headings, Part titles
Section Title	Cormorant Garamond	2–3.2rem / 300	Section h2 headings
UI Label	DM Mono	0.65–0.75rem / 400, uppercase, tracked	Eyebrows, part numbers, tags
Body	DM Sans	0.9–1rem / 300	All paragraph content
Code / Mono	DM Mono	0.8rem / 400	Code blocks, clinical formulas
Chapter Body (docs)	DM Sans	1rem / 300–400	MDX content pages

3.3 Spacing & Grid

- Base unit: 8px
- Section padding: 6rem 3rem (desktop), 4rem 1.5rem (mobile)
- Nav height: 64px (fixed)
- Max content width: 1200px (landing), 860px (docs pages)
- Grid gap (part cards): 1px on background rgba(gold, 0.08) — creates hairline dividers

3.4 Component Inventory

Component	File	Description
`<AbcdeBox />`	components/AbcdeBox.jsx	Structured ABCDE assessment template with expandable rows
`<KeyPoints />`	components/KeyPoints.jsx	Gold-bordered callout for chapter key points
`<FigurePlaceholder />`	components/ FigurePlaceholder.jsx	Labelled placeholder with figure number and caption slot
`<SampleHistory />`	components/ SampleHistory.jsx	SAMPLE history mnemonic structured display
`<AlgoCard />`	components/AlgoCard.jsx	Step-based algorithm card (future: interactive)
`<ClinicalTable />`	components/ ClinicalTable.jsx	Styled table component with navy header row
`<PartBadge />`	components/PartBadge.jsx	Part number + title badge for cross-references

4. Project Structure

The full directory tree of the Docusaurus project:

```
css-handbook/
├── docs/                          # All MDX content files
│   ├── index.md                   # Docs landing / intro page
│   ├── part1-deteriorating-patient/
│   │   ├── _category_.json        # Sidebar label config
│   │   ├── 1-1-abcde-sample.mdx
│   │   ├── 1-2-ecg-interpretation.mdx
│   │   ├── 1-3-vascular-access.mdx
│   │   └── 1-4-als-algorithm.mdx
│   ├── part2-cardiac-emergencies/
│   │   ├── _category_.json
│   │   ├── 2-1-acs.mdx
│   │   ├── 2-2-arrhythmias.mdx
│   │   └── 2-3-ecg-emergency.mdx
│   ├── part3-respiratory/
│   ├── part4-neurological/
│   ├── part5-sepsis/
│   ├── part6-fluids-electrolytes/
│   └── part7-injuries/
├── src/
│   ├── components/                # Reusable React components
│   │   ├── AbcdeBox.jsx
│   │   ├── KeyPoints.jsx
│   │   ├── FigurePlaceholder.jsx
│   │   ├── SampleHistory.jsx
│   │   └── ClinicalTable.jsx
│   ├── css/
│   │   └── custom.css             # All CSS overrides + design tokens
│   └── pages/
│       └── index.js               # Landing page (from landing.html)
├── static/
│   ├── img/
│   │   ├── logo.svg
│   │   └── figures/               # Chapter screenshots (added per phase)
│   └── _redirects                 # Netlify redirect rules
├── docusaurus.config.js           # Main config
├── sidebars.js                    # Sidebar structure
└── package.json
```

4.1 Category Config Files

Each part folder contains a _category_.json that controls sidebar display:

```
// docs/part1-deteriorating-patient/_category_.json
{
  "label": "Part 1 · Deteriorating Patient",
  "position": 1,
  "collapsible": true,
  "collapsed": false,
  "className": "sidebar-part"
}
```

4.2 MDX Chapter Template

Every content chapter follows this structure:

```
---
title: ABCDE & SAMPLE Assessment
sidebar_label: 1.1 ABCDE & SAMPLE
sidebar_position: 1
description: Structured approach to initial assessment of deteriorating patients
tags: [assessment, abcde, sample, deteriorating-patient]
---


import KeyPoints from "@site/src/components/KeyPoints";
import FigurePlaceholder from "@site/src/components/FigurePlaceholder";
import AbcdeBox from "@site/src/components/AbcdeBox";


# ABCDE & SAMPLE Assessment


<KeyPoints items={[
  "Apply ABCDE systematically before any intervention",
  "Treat life-threatening findings at each step before moving on",
  "Re-assess after every intervention"
]} />


## Introduction


Body text goes here...


<FigurePlaceholder number="1.1" caption="ABCDE assessment flowchart" />


## The ABCDE Approach


<AbcdeBox />
```

5. Content Pipeline

The handbook PDF has 108 pages of extractable text, but the raw extraction contains encoding artifacts (special characters rendered as !, &, %, numbers). Every chapter therefore requires a human review pass before publishing. This section defines the workflow.

5.1 Extraction Workflow per Chapter

1. **Extract raw text:** Use pdfplumber (Python) to extract text from the target page range for the chapter.
2. **Identify encoding corruption:** Scan for ! & % and digit-only tokens that should be special characters (e.g., č, š, ž, ć, đ in Croatian names; ≥ ≤ → in clinical content).
3. **Manual clean pass:** Fix corrupted characters, reformat tables, restore list structure, confirm all clinical values (doses, thresholds) against the print PDF.
4. **Insert component placeholders:** Replace every figure/diagram reference with a <FigurePlaceholder /> tag, noting the correct figure number and caption.
5. **Paste into .mdx file:** Add front matter, imports, and cleaned content. Run locally (npm start) to verify rendering.
6. **Screenshot figures (async):** When ready, take high-resolution screenshots of the relevant PDF pages, export as PNG, place in /static/img/figures/, and replace placeholders.

CRITICAL QA RULE

Never publish a chapter without verifying all clinical values against the source PDF.

Drug doses, threshold values (e.g. SpO2 <94%, GCS <9), and classification criteria must be exact.

Typography fixes (quotes, dashes) can be done in batch — clinical accuracy cannot.

5.2 Chapter Map & Status

Chapter	Part	Pages (PDF)	MDX File	Status
1.1 ABCDE & SAMPLE	Part 1	15–28	1-1-abcde-sample.mdx	Pending
1.2 ECG Interpretation	Part 1	29–42	1-2-ecg-interpretation.mdx	Pending
1.3 Vascular Access	Part 1	43–50	1-3-vascular-access.mdx	Pending

Chapter	Part	Pages (PDF)	MDX File	Status
1.4 ALS Algorithm	Part 1	51–58	1-4-als-algorithm.mdx	Pending
2.1 Acute Coronary Syndrome	Part 2	59–66	2-1-acs.mdx	Pending
2.2 Cardiac Arrhythmias	Part 2	67–78	2-2-arrhythmias.mdx	Pending
3.1 Acute Respiratory Failure	Part 3	81–90	3-1-resp-failure.mdx	Pending
3.2 Oxygen & Pulse Oximetry	Part 3	87–90	3-2-oxygen.mdx	Pending
3.3 Arterial Blood Gases	Part 3	91–96	3-3-abg.mdx	Pending
3.4 Anaphylaxis	Part 3	97–102	3-4-anaphylaxis.mdx	Pending
3.5 Acute Bronchospasm	Part 3	103–112	3-5-bronchospasm.mdx	Pending
3.6 Pulmonary Oedema	Part 3	113–118	3-6-pulm-oedema.mdx	Pending
4.1 Neurological Deterioration	Part 4	127–134	4-1-neuro.mdx	Pending
4.2 Seizures	Part 4	135–140	4-2-seizures.mdx	Pending
4.3 Acute Poisoning	Part 4	141–145	4-3-poisoning.mdx	Pending
5.1 Sepsis & Septic Shock	Part 5	147–153	5-1-sepsis.mdx	Pending
6.1 IV Fluids	Part 6	155–160	6-1-iv-fluids.mdx	Pending
6.2 Electrolyte Disorders	Part 6	161–172	6-2-electrolytes.mdx	Pending
6.3 Haemorrhage & Shock	Part 6	173–181	6-3-haemorrhage.mdx	Pending
7.1 Trauma Assessment	Part 7	183–204	7-1-trauma.mdx	

Chapter	Part	Pages (PDF)	MDX File	Status
				Pending
7.2 Burns	Part 7	205–212	7-2-burns.mdx	Pending

6. Docusaurus Configuration

6.1 docusaurus.config.js

```
// docusaurus.config.js
const config = {
  title: 'Kabinet Vještina',
  tagline: 'Simulation of Clinical Skills — A Practical Guide',
  favicon: 'img/favicon.ico',
  url: 'https://css-handbook.netlify.app',
  baseUrl: '/',
  organizationName: 'medri-rijeka',
  projectName: 'css-handbook',
  onBrokenLinks: 'throw',
  onBrokenMarkdownLinks: 'warn',

  i18n: { defaultLocale: 'en', locales: ['en'] },

  presets: [['classic', {
    docs: {
      sidebarPath: require.resolve('./sidebars.js'),
      routeBasePath: 'handbook',  // /handbook/ not /docs/
    },
    blog: false,
    theme: { customCss: require.resolve('./src/css/custom.css') },
  }]],

  plugins: ['@docusaurus/plugin-search-local'],

  themeConfig: {
    navbar: {
      title: 'Kabinet Vještina',
      logo: { alt: 'KV Logo', src: 'img/logo.svg' },
      items: [
        { to: '/handbook', label: 'Handbook', position: 'left' },
        { to: '/about', label: 'About', position: 'left' },
        { href: 'https://github.com/...', label: 'GitHub', position: 'right' },
      ],
    },
    colorMode: { defaultMode: 'dark', disableSwitch: false },
  },
};
module.exports = config;
```

6.2 Custom CSS Tokens

```
/* src/css/custom.css */
:root {
  --ifm-color-primary:         #c8a96e;
  --ifm-color-primary-dark:    #b8943a;
  --ifm-color-primary-darker:  #a07c28;
  --ifm-color-primary-darkest: #7a5f1c;
  --ifm-color-primary-light:   #d8bc8a;
  --ifm-color-primary-lighter: #e2c98a;
  --ifm-color-primary-lightest:#f0deb0;
  --ifm-background-color:      #0b1120;
  --ifm-navbar-background-color: rgba(11,17,32,0.9);
```

```
  --ifm-font-family-base:      'DM Sans', sans-serif;
  --ifm-font-family-monospace: 'DM Mono', monospace;
  --ifm-heading-font-family:   'Cormorant Garamond', serif;
  --ifm-font-size-base:        1rem;
  --ifm-line-height-base:      1.75;
  --kv-navy:  #0b1120;
  --kv-gold:  #c8a96e;
  --kv-cream: #f2ede4;
}
```

7. Component Specifications

7.1 FigurePlaceholder

Renders a visually consistent placeholder wherever a figure/diagram belongs, with figure number and caption. Replace by swapping src prop once screenshots are ready.

```
// src/components/FigurePlaceholder.jsx
export default function FigurePlaceholder({ number, caption, src }) {
  if (src) return (
    <figure className='kv-figure'>
      <img src={src} alt={caption} />
      <figcaption>Figure {number}. {caption}</figcaption>
    </figure>
  );
  return (
    <div className='kv-figure-placeholder'>
      <div className='kv-figure-placeholder__inner'>
        <span className='kv-figure-placeholder__label'>
          Figure {number}
        </span>
        <span className='kv-figure-placeholder__caption'>{caption}</span>
      </div>
    </div>
  );
}
```

7.2 KeyPoints

```
// src/components/KeyPoints.jsx
export default function KeyPoints({ items }) {
  return (
    <div className='kv-keypoints'>
      <p className='kv-keypoints__label'>Key Points</p>
      <ul>
        {items.map((item, i) => <li key={i}>{item}</li>)}
      </ul>
    </div>
  );
}
```

7.3 AbcdeBox

Renders the standard ABCDE assessment framework as a structured table with expandable detail rows per letter. Pre-populated with default clinical content; accepts an override prop for chapter-specific versions.

8. Deployment

8.1 Netlify (Recommended)

7. Connect GitHub repository to Netlify (netlify.com → New Site → Import from Git)
8. Build command: npm run build
9. Publish directory: build
10. Node version: 18 (set in netlify.toml or environment variables)
11. Deploy triggers on every push to main branch automatically

```
# netlify.toml
[build]
  command   = "npm run build"
  publish   = "build"

[build.environment]
  NODE_VERSION = "18"
```

8.2 GitHub Pages

```
# package.json — add deploy scripts:
"scripts": {
  "deploy": "docusaurus deploy",
  "build":  "docusaurus build"
},
"docusaurus": {
  "organizationName": "your-org",
  "projectName": "css-handbook"
}
```

8.3 Custom Domain

To use a faculty domain (e.g. vjestina.medri.uniri.hr): add a CNAME record pointing to the Netlify subdomain, then configure the custom domain in Netlify dashboard. SSL is provisioned automatically.

9. Build Phases & Milestones

Phase	Title	Key Deliverables	Owner
P0	Scaffold	Docusaurus project init, custom theme (CSS tokens), landing page as homepage, empty chapter stubs for all 21 chapters, deploy to Netlify	Dev
P1	Part 1	Extract + clean all Part 1 chapters (1.1–1.4), all reusable components built, figure placeholders in place, local search enabled	Dev + Content
P2	Parts 2–3	Cardiac Emergencies + Respiratory Emergencies chapters live, ClinicalTable and AlgoCard components added	Dev + Content
P3	Parts 4–5	Neurological Deterioration + Sepsis chapters live	Content
P4	Parts 6–7	Fluids/Electrolytes + Injuries chapters live — full handbook published	Content
P5	Figures	Replace all FigurePlaceholder instances with actual screenshots, add alt text	Content
P6	Polish	Algolia search, mobile QA, print stylesheet, accessibility audit, analytics	Dev

10. Local Development

```
# Initial setup
npx create-docusaurus@latest css-handbook classic --typescript
cd css-handbook

# Install extras
npm install @docusaurus/plugin-search-local

# Start dev server (hot reload)
npm start
# → opens http://localhost:3000

# Production build (test before deploy)
npm run build
npm run serve
```

```
# Deploy to GitHub Pages
npm run deploy
```

DEV ENVIRONMENT NOTE

Docusaurus requires Node.js >= 18.0.

Recommended: use nvm to manage Node versions.

Windows users: WSL2 is strongly recommended over native Windows for Docusaurus.

Run npm start — the dev server auto-reloads on any .mdx or .jsx file change.

11. Conventions & Rules

File Naming

- All MDX files: kebab-case, prefixed with chapter number: 1-1-abcde-sample.mdx
- All components: PascalCase: KeyPoints.jsx, FigurePlaceholder.jsx
- All images: kebab-case with chapter prefix: fig-1-1-abcde-flowchart.png

Content Rules

- All clinical values must be verified against the source PDF before merge
- Every chapter must have at minimum: front matter, one <KeyPoints />, section headings matching the book
- Never remove a <FigurePlaceholder /> without replacing it with the actual figure
- Croatian names and medical terms: verify diacritics manually (PDF extraction corrupts them)

Git Workflow

- main branch: always deployable
- Feature branches: content/part-1-chapter-1, feat/algo-card-component, fix/ecg-encoding
- One PR per chapter or component — keeps review scope manageable

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